

ORIGINAL ARTICLE**Harms were detected but not reported in six clinical trials of gabapentin**Evan Mayo-Wilson^{a,b,*}, Riaz Qureshi^c, Hwanhee Hong^d, Xiwei Chen^b, Tianjing Li^c^aDepartment of Epidemiology, UNC Gillings School of Global Public Health, Chapel Hill, NC, USA^bDepartment of Epidemiology and Biostatistics, Indiana University School of Public Health-Bloomington, Bloomington, IN, USA^cDepartment of Ophthalmology, School of Medicine, University of Colorado Anschutz Medical Campus, Aurora, CO, USA^dDepartment of Biostatistics and Bioinformatics, Duke University School of Medicine, Durham, NC, USA

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Abstract**Objectives:** We sought to assess and report harms that were observed but not disclosed previously in clinical trials of gabapentin.**Study Design and Setting:** We reanalyzed individual participant data from six randomized parallel trials of gabapentin for neuropathic pain, and we conducted meta-analyses. Between 1996 and 2003, adult participants were assigned to gabapentin (600 mg–3,600 mg per day) or placebo for 7–14 weeks. We calculated the proportion of participants with: one or more adverse events (AEs), one or more serious AEs, discontinued, discontinued because AEs. We also estimated effects on AEs at three levels of aggregation using COSTART, a hierarchical system for classifying AEs: body system, midlevel system, preferred term.**Results:** We found evidence of important harms that were neither included in published trial reports nor included in systematic reviews. Aggregating related harms led to greater confidence that gabapentin might harm the nervous system and possibly the digestive, metabolic and nutritional, respiratory, sensory, and urogenital body systems. Nervous system harms were more common than previous reports suggest.**Conclusion:** Clinical trials identified harms that were not reported publicly, and journal articles overstated uncertainty about harms. Relying on journal articles to evaluate gabapentin's harms could contribute to incomplete and misleading conclusions in systematic reviews and guidelines. To prevent bias arising from the selective nonreporting of results, journal articles should describe how to access data for all harms observed in clinical trials (e.g., by sharing individual participant data). © 2023 Elsevier Inc. All rights reserved.**Keywords:** Neuropathic pain; Gabapentinoids; Adverse events; Harms; Clinical trials; Systematic reviews; Meta-analysis**1. Introduction**

In 1993, gabapentin (Neurontin®) was approved to treat epilepsy by the U.S. Food and Drug Administration (FDA). To increase sales, gabapentin's manufacturer encouraged

“off label” prescribing by publishing favorable clinical trial results in medical journals [1–3]. Those articles overestimated gabapentin's benefits and underestimated its harms [4–10]. Pfizer pleaded guilty to criminal and civil charges for illegal marketing, and it settled claims of defrauding patients and benefit providers [11,12].

Today, gabapentin continues to be prescribed widely for multiple conditions [13–16]. Prescribing might be influenced by systematic reviews and guidance based on incomplete evidence. For example, a Cochrane systematic review of gabapentin summarizes evidence of harms reported in journal articles, but it does not include evidence from other reports of the included trials [17,18]. Summaries in *JAMA* [19] and *BMJ* [20] likewise rely on evidence about harms reported in journal articles.

More information about harms that occurred in six early gabapentin trials is available because individual participant data (IPD) and clinical study reports (CSRs) [21] became available through litigation against Pfizer. These early trials influenced regulatory decisions, clinical practice, and later

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Ethical approval: This study was reviewed by the Indiana University Human Research Protection Program and determined to be not human subjects research and thus exempt from institutional review board review (protocol #1910607198).

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